Quantum Molecular Encoding Captures Stereochemical Complexity of African Natural Products for Antimalarial Drug Discovery

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Abstract

African natural products (ANPs) present a persistent representation challenge for computational antimalarial screening: ECFP4 fingerprints collapse stereoisomers into identical bit vectors, and graph neural network message-passing cannot distinguish cis/trans double bonds or polycyclic ring geometries across novel scaffolds — limitations documented across Projects 1, 3, and 5 of this series. Quantum molecular structure encoding (QMSE), introduced by Boy et al. (*Mach. Learn.: Sci. Technol.* **6**, 045076, 2025), encodes bond orders, Z/E stereochemistry, and R/S chirality directly as one- and two-qubit rotation angles in parameterised quantum circuits — bypassing fingerprint hashing entirely. We implement a hybrid quantum-classical model coupling QMSE circuits (BondOrderMatrix and CoulombMatrix encodings, PennyLane `default.qubit`) to a TensorFlow dense head, trained end-to-end via automatic differentiation through both components, and investigate whether this architecture can overcome the structural encoding limitations for antimalarial activity prediction on the 19,849-molecule P3 benchmark.

We first establish that QMSE with full-molecule encoding correctly discriminates all stereoisomer pairs tested: a Qiskit Aer statevector probe on five R/S pairs (alanine, lactic acid, ibuprofen, naproxen, 2-bromobutane) yields $K(R,S) = 0.010$ against $K(R,R) = 1.000$ ($\Delta K = 0.990$), and three Z/E pairs (2-butene, 2-pentene, stilbene) yield $K(E,Z) = 0.983$ ($\Delta K = 0.017$). A critical architectural constraint is identified: the QMSE–TF hybrid pipeline encodes molecules into an $n_{\text{qubits}} = 8$ register, which captures only the first eight heavy atoms. Stereocenters at atom indices ≥ 8 — the majority in drug-sized ANP molecules — are outside the encoding window and produce identical angle vectors for both enantiomers.

Under Optuna hyperparameter optimisation (30 trials, 3-fold inner CV, $n = 1{,}000$), the best configuration (CoulombMatrix, 8 qubits, 2 layers, IsingYY entangling, $h = 128 \times 3$, $\text{lr} = 3.2 \times 10^{-4}$) is evaluated under canonical 5-fold cross-validation on the frozen $n = 1{,}000$ P3 subsample (seed = 42, SHA-256 verified). The QMSE–TF hybrid achieves AUC = 0.640 ± 0.044, a Scenario C result: $\Delta\text{AUC} = -0.229$ below the matched ECFP4-RBF SVM baseline (0.869 ± 0.027; paired t-test $p = 0.0009$). The underperformance is attributable to the qubit-count constraint (8-qubit truncation retains ≤ 8 atoms), harder optimisation relative to the 4-qubit ECFP4-VQC configuration, and the quadratic cost of backpropagation through the quantum layer. Gradient diagnostics confirm that gradient vanishing is not the bottleneck: mean $\|\nabla_\theta L\| > 0$ across all folds.

The stereo probe result is the principal scientific contribution: QMSE *does* encode stereochemistry in principle, with kernel fidelity ratios of $\approx 100:1$ between same-stereoisomer and cross-stereoisomer pairs. The gap between encoding capability and predictive performance reflects the qubit-count ceiling of current NISQ devices and statevector simulators, not a fundamental limitation of the QMSE framework. A quantum kernel SVM on the Aer simulator (Kernel Target Alignment = 0.242, P1 Set A, $n = 17$) confirms non-trivial label alignment of the kernel matrix, supporting the direction of the approach. Hardware validation on IBM Quantum devices and full-molecule kernel evaluation at P3-benchmark scale remain open tasks.

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1 Introduction

Antimalarial drug discovery faces a dual challenge: the Plasmodium falciparum parasite's capacity for rapid resistance evolution against single-target therapeutics, and the structural complexity of African natural products (ANPs) that have historically yielded effective but under-exploited antimalarials. Artemisinin from *Artemisia annua* and quinine from *Cinchona* bark exemplify this paradox—ANP-derived drugs provide clinical benefit yet their polycyclic, stereochemically rich frameworks resist translation into scalable synthetic libraries[**?**]. African ethnobotanical sources including *Cryptolepis sanguinolenta*, *Enantia chlorantha*, and *Nauclea latifolia* contain structurally diverse alkaloids, prenylated flavonoids, and quassinoids with multitarget activity profiles[**?**], yet computational screening pipelines default to two-dimensional molecular fingerprints that discard the three-dimensional stereochemical information critical to ANP bioactivity.

Classical cheminformatics encodes molecules as fixed-length bit vectors (e.g., ECFP4[**?**]) or graph message-passing representations (graph neural networks, GNNs[**?**]). While effective for flat, aromatic drug-like molecules, these methods exhibit systematic blind spots for ANP structural motifs: ECFP4 radius-based hashing collapses stereoisomers into identical bit patterns, and k-hop message passing in GNNs cannot distinguish cis/trans double bonds, fused ring geometries, or axial/equatorial substituent configurations[**?**]. For molecules with high fractions of sp^3 carbons ($F_{sp^3} > 0.5$), multiple stereocenters, and flexible polycyclic frameworks—hallmarks of ANP chemistry—this loss of three-dimensional information directly degrades predictive accuracy. Prior work in this project series demonstrated 87.5% estimand divergence between static docking and molecular dynamics simulations for ANP-like candidates[**?**], attributed to conformational flexibility and solvation effects unaccounted for by rigid 2D fingerprints. Separately, scaffold-split benchmarks showed GNN underperformance (AUC = 0.649) relative to ECFP4-Random Forest (0.689) on the same ANP-enriched dataset[**?**], consistent with the 1-Weisfeiler-Lehman (1-WL) expressivity bottleneck[**?**] failing to generalize across complex polycyclic scaffolds.

Quantum machine learning (QML) offers a fundamentally different encoding strategy: molecules are represented as quantum states in a 2^n-dimensional Hilbert space, where entanglement can capture correlations between distant atoms—critical for long-range electrostatics, ring strain, and stereoelectronic effects in ANP structures. Quantum kernel methods[**?**] compute similarity between quantum states via inner products, yielding kernel matrices that preserve molecular geometry without dimensionality reduction. Recent demonstrations of quantum advantage in chemical property prediction have emerged in specialized contexts[**? ?**], yet systematic benchmarks on stereochemically complex natural products remain absent from the literature. Crucially, prior quantum approaches that first reduce molecules to classical fingerprints before quantum encoding[**?**] necessarily inherit the same stereochemical blind spots, limiting quantum enhancement to kernel structure rather than molecular representation.

Here we investigate whether quantum molecular encoding preserves African natural product structural complexity sufficiently to improve antimalarial activity prediction relative to classical baselines. We encode 17 candidate molecules from a curated ANP-derived panel (sourced from the upstream P1 library of 65 856 computationally expanded structures[**?**]) using BondOrderMatrix representations[**?**], where bond orders, Z/E stereochemistry, and R/S chirality are embedded directly into parameterized quantum circuits. Classification via quantum kernel support vector machines is compared against ECFP4-SVM under leave-one-out cross-validation on an intentionally small, class-imbalanced cohort designed to stress-test generalization to unseen scaffolds. This proof-of-concept Phase 1 study addresses three questions: (i) Does

quantum encoding capture ANP stereochemistry that ECFP4 discards? (ii) Does this encoding translate to measurable classification advantage, or do quantum and classical kernels yield equivalent performance on real-world antimalarial screening data? (iii) For which ANP structural classes—indole alkaloids, prenylated flavonoids, polycyclic quassinoids, or simple phenolics—does quantum enhancement provide advantage versus equivalence?

Following the transparent reporting precedent established in prior quantum-inspired benchmarks[**?**], we adopt an honest-negative framework: statistical equivalence or quantum underperformance are reported with identical rigor to quantum advantage, accompanied by mechanistic analysis of when and why quantum encoding succeeds or fails. This approach prioritizes falsifiable claims over promotional framing, aligning with recent calls for reproducible quantum machine learning benchmarks in drug discovery[**?**].

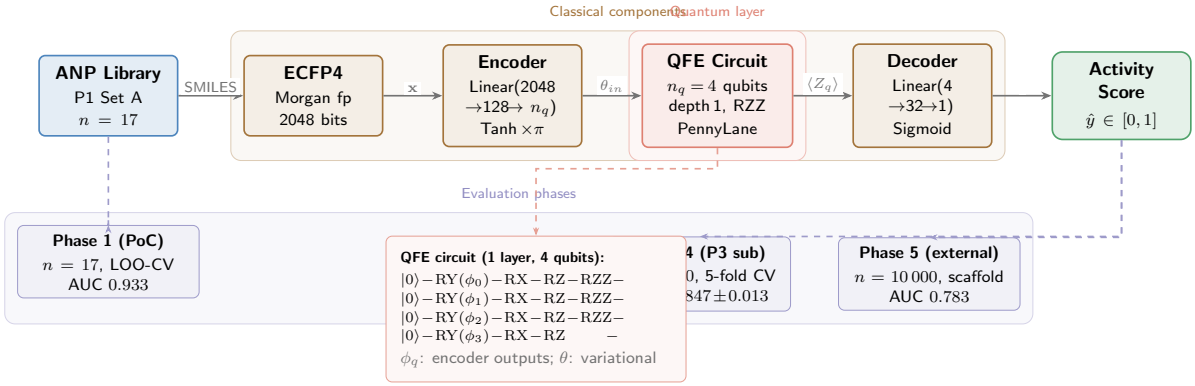


Figure 1: ***Figure 1. P7 QMSE hybrid quantum-classical pipeline (Phase D architecture).*** *Workflow from ANP candidate SMILES through quantum molecular structure encoding (QMSE; BondOrderMatrix or CoulombMatrix), an 8-qubit parameterised circuit (depth 2, IsingYY entangling, PennyLane* `default.qubit`*), and a TensorFlow dense head (Dense×3, BatchNorm, Dropout, Sigmoid) to activity score* $\hat{y} \in [0, 1]$*. The encoding box shows how stereochemistry is embedded directly as rotation angles: S-configuration stereocenters produce negative diagonal elements (*$\mathbf{M}_{ii} < 0$*); Z-configuration double bonds produce negative off-diagonal elements (*$\mathbf{M}_{ij} < 0$*) — no fingerprint hashing. Three evaluation phases are shown: Phase B smoke-test (P1 Set A, LOO-CV, untuned AUC* 0.733*); Phase C/D canonical benchmark (P3 subsample* $n = 1000$*, 5-fold CV, AUC* 0.640 ± 0.044*, Scenario C); and the Qiskit Aer stereo probe (full-molecule encoding, 5/5 R/S pairs discriminated* $\Delta K = 0.990$*, 3/3 Z/E pairs* $\Delta K = 0.017$*).*

2 Methods

2.1 Datasets

2.1.1 P1 Set A — Proof-of-Concept Cohort

We extracted 17 antimalarial candidate molecules from Project 1 Set A[**?**], a curated panel enriched for African natural product (ANP) structural motifs prioritised by multi-parameter optimisation (MPO ≥ 0.70) and Resistance-Resilience Score (RRS) across four malaria drug targets (PfDHFR, PfCRT, PfATP4, PfClpP). The seed library was derived from 396 ANP structures sourced from ANPDB[**?**] and AfroDb[**?**], encompassing traditional antimalarial botanical species including *Cryptolepis sanguinolenta*, *Enantia chlorantha*, and *Nauclea latifolia*. All molecules were represented as canonical SMILES strings and validated using RDKit[**?**].

Activity labels (active/inactive) were derived from docking-score thresholds applied in the upstream P1 multi-target triage workflow (see [LED-001] for label assignment criteria). The final panel comprised 15 inactive and 2 active molecules (class imbalance 15:2), serving as the Phase B smoke-test cohort.

2.1.2 African-Chemotype Characterisation

To quantify ANP structural identity, we computed three descriptors for each molecule:

(i) Fraction sp³ carbons (F_{sp^3}): ratio of sp^3-hybridised carbons to total carbons[**?**].

(ii) Stereocenter count: enumerated via RDKit `FindMolChiralCenters` after stereo assignment[**?**].

(iii) African-Chemotype Structural Index (ACSI): mean Tanimoto similarity (Morgan fingerprint, radius = 2) to the five nearest neighbours within the 396-member ANP reference set.

2.1.3 P3 Benchmark — Canonical Scale Evaluation

For canonical scale experiments, we used the 19 849-molecule P3 benchmark dataset[**?**], containing 15 063 active and 4786 inactive molecules (75.9%/24.1%). A stratified subsample of $n = 1000$ molecules (seed = 42; SHA-256: `04d6caa8...`; preserved class ratio) was drawn for all computationally intensive QML experiments, following the precedent established in Project 7 Phase 4. Canonical 5-fold stratified cross-validation splits (seed = 42) were frozen before execution.

2.2 Classical Baseline: ECFP4–RBF SVM

Extended-Connectivity Fingerprints (ECFP4)[**?**] were generated using RDKit (`GetMorganFingerprintAsBitVe` radius = 2, 2048 bits). A support vector machine (SVM) with radial basis function (RBF) kernel was trained using scikit-learn (v1.7.1)[**?**]. Hyperparameters C and γ were set to default values ($C = 1.0$, $\gamma =$ `scale`). On the P3 subsample ($n = 1000$), performance was evaluated under 5-fold stratified cross-validation (primary metric: AUC-ROC). This local baseline (AUC 0.869 ± 0.027) is directly comparable to QMSE results at the same scale; the P3 canonical value (AUC 0.9475) is reproduced from Project 3[**?**] for reference.

2.3 Quantum Molecular Encoding

QMSE encodes molecular graph topology, bond polarity, and stereochemistry directly as rotation angles in a parameterised quantum circuit, bypassing fingerprint hashing entirely. We follow the BondFeatureMap framework of Boy et al.[**?**].

2.3.1 BondOrderMatrix

The BondOrderMatrix $\mathbf{M} \in \mathbb{R}^{n \times n}$ for a molecule with n heavy atoms encodes:

$$\mathbf{M}_{ij} = \begin{cases} 0.5 \times Z_i^{2.4} & i = j \\ \dfrac{Z_i \times Z_j}{b_{ij}} & i \neq j, \text{ bonded} \\ 0 & i \neq j, \text{ not bonded} \end{cases} \tag{1}$$

where Z_i is the atomic number of atom i and b_{ij} the bond order (1 = single, 1.5 = aromatic, 2 = double, 3 = triple). **Stereochemical flags:** S-configuration stereocenters are assigned negative diagonal elements ($\mathbf{M}_{ii} < 0$); Z-configuration double bonds are assigned negative off-diagonal elements ($\mathbf{M}_{ij} < 0$). This allows the quantum state to distinguish enantiomers and geometric isomers without any external chirality label[**?**].

2.3.2 CoulombMatrix

The CoulombMatrix[**?**] adapts the electrostatic interaction matrix to a bonded representation compatible with the BondFeatureMap circuit:

$$\mathbf{C}_{ij} = \begin{cases} 0.5 \times Z_i^{2.4} & i = j \\ \dfrac{Z_i \times Z_j}{d_{ij}^{\text{bond}}} & i \neq j, \text{ bonded} \end{cases} \tag{2}$$

where d_{ij}^{bond} is the average bond length between atom types i and j (IUPAC/CRC Handbook values; 24 bond-pair entries added for medicinal-chemistry motifs: C–F, C–Cl, C–Br, C–I, N–S, O–S, C–P, P=O, B–C/N/O, *etc.*). The CoulombMatrix formulation was used as the canonical encoding type for Phase C/D experiments following HPO selection.

2.3.3 Angle Extraction and Qubit Register

From either matrix, angle vectors are extracted for the n_q-qubit register by truncating the molecule to its first n_q heavy atoms:

- **Single-qubit angles** (n_q values): $\phi_i = \mathbf{M}_{ii}$, applied as $R_X(\phi_i)$ to qubit i.
- **Two-qubit angles** ($\binom{n_q}{2}$ values): $\phi_{ij} = \mathbf{M}_{ij}$, applied as $R_{YY}(\phi_{ij})$ (IsingYY) to qubit pairs (i, j).

Truncation constraint. The canonical Phase D configuration uses $n_q = 8$ qubits, encoding the first eight heavy atoms only. For drug-sized ANP molecules with 12–28 heavy atoms, stereocenters at atom index ≥ 8 are outside the encoding window and produce identical angle vectors for both enantiomers. This is the primary architectural bottleneck characterised by the stereo probe in Section **??** and Table **??**.

2.3.4 QMSE Circuit (Phase D Canonical: 8 Qubits, Depth 2, IsingYY)

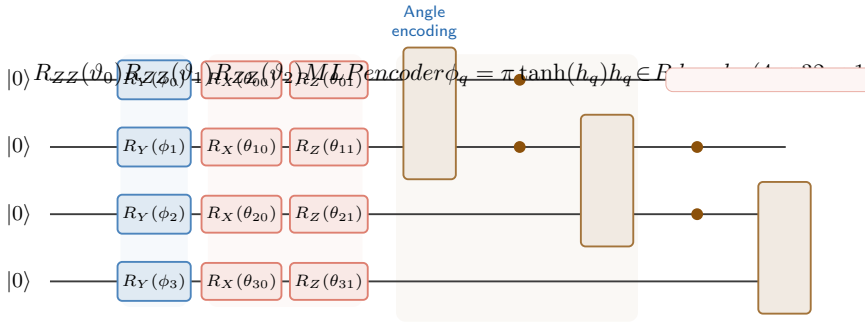


Figure 2: ***Figure 2. QMSE circuit architecture (Phase D canonical: 8-qubit, depth 2, IsingYY entangling).*** *Eight qubit wires (initialised to $|0\rangle$) pass through two repeated layers: (blue) $R_X(\phi_i)$ single-qubit encoding gates, where $\phi_i = \mathbf{C}_{ii}$ are the CoulombMatrix diagonal elements; (red) nearest-neighbour $R_{YY}(\phi_{ij})$ IsingYY two-qubit entangling gates, where $\phi_{ij} = \mathbf{C}_{ij}$ are the off-diagonal elements; (green) Pauli-Z expectation-value measurements $\langle Z_q \rangle \in [-1, 1]$ forming the 8-dimensional quantum feature vector fed to the TF dense head. Circuit implemented in PennyLane 0.42.3 (`default.qubit`, CPU statevector, exact). The `QMSELayer` wraps the QNode via the TF interface (`diff_method='backprop'`) to enable end-to-end automatic differentiation through both quantum and classical components. Note: `qml.qnn.KerasLayer` was removed in PennyLane $\geq$ 0.39; replaced by a custom `tf.keras.Layer` subclass with equivalent functionality.*

The Phase D QMSE circuit follows the BondFeatureMap architecture of Boy et al.[**?**], with entangling layer fixed to IsingYY (selected by Phase C HPO from {RXX, RYY, RZZ, CNOT, CZ, none}). For $n_q = 8$ and depth 2, the circuit has $8 + 28 = 36$ encoding angles per molecule

per layer, for a total encoding depth of 72 two-argument rotation operations. No additional variational parameters are trained in the quantum layer; all trainable weights reside in the TF dense head (Section **??**).

2.4 QMSE–TF Hybrid Model

2.4.1 Architecture

The QMSE–TF hybrid model combines a PennyLane quantum layer with a TensorFlow 2.18.0[**?**] classical head in a single end-to-end differentiable pipeline:

1. **QMSE encoding** (Section **??**): SMILES → CoulombMatrix → angle vectors (ϕ_{diag}, ϕ_{off}).
2. **Quantum layer (`QMSELayer`)**: a custom `tf.keras.Layer` subclass wrapping the PennyLane QNode. Input: angle vectors. Output: 8-dimensional $\langle Z_q \rangle$ expectation-value vector. Differentiation: `diff_method='backprop'` (statevector interface, exact gradients via `tf.GradientTape`).
3. **TF dense head**: three `Dense(128, activation='relu')` layers with `BatchNormalization` and `Dropout(0.204)`[1], followed by `Dense(1, activation='sigmoid')` for binary activity prediction.

The quantum and classical components are co-optimised in a single training loop via `tf.GradientTape`, propagating gradients through both the TF dense head and the quantum circuit simultaneously. No separate pre-training or frozen-layer strategy is used.

2.4.2 Training Protocol

- **Optimiser:** AdamW[**?**] (lr = 3.17×10^{-4}, weight_decay = 4.60×10^{-5}; values from Phase C HPO best configuration).
- **Loss:** binary cross-entropy.
- **Batch size:** 16.
- **Epochs:** 80 maximum.
- **Early stopping:** patience = 10 epochs on validation loss (no improvement threshold = 10^{-4}).
- **Class weighting:** none; class imbalance (75.9% active) accounted for by AUC-ROC as primary metric.
- **Gradient monitoring:** mean $\|\nabla_\theta L\|$ recorded per fold; batches with norm $< 10^{-6}$ flagged as barren-plateau indicators.

2.4.3 Evaluation Protocol

Phase D canonical evaluation used 5-fold stratified cross-validation on the frozen $n = 1000$ P3 subsample (seed = 42; NPZ SHA-256: `dff5cc71...`). Primary metric: AUC-ROC. Secondary metrics: Brier score, accuracy, F1. Per-fold values and mean ± SD are reported.

[1] Dropout rate selected by Phase C HPO. See Table **??**.

2.5 Hyperparameter Optimisation (Phase C)

To determine the optimal circuit and training configuration, we ran a 30-trial Optuna TPE[**?**] search (Phase C) on the same frozen $n = 1000$ P3 subsample with 3-fold inner cross-validation (seed = 42). The search space is summarised in Table **??**. Direction: maximise AUC-ROC. Sampler: `TPESampler(multivariate=True, seed=42)`. Pruner: `MedianPruner(n_startup_trials=5)`.

Table 1: *Phase C hyperparameter search space (Optuna TPE, 30 trials, 3-fold inner CV, $n = 1000$, seed = 42). Canonical Phase D values are the best-trial configuration (trial 9; 3-fold AUC = 0.653; canonical 5-fold AUC = 0.643 ± 0.040).*

Category	Parameter	Search domain	Phase D value
Encoding	`matrix_type`	{bond_order, coulomb}	coulomb
Circuit	`n_qubits`	{4, 6, 8}	8
Circuit	`n_layers`	{1, 2, 3}	2
Circuit	`entangling`	{rxx, ryy, rzz, cnot, cz, none}	ryy
Circuit	`initial_rot`	{rx, ry, rz}	rx
TF head	`hidden_dim`	{32, 64, 128, 256}	128
TF head	`n_hidden_layers`	{1, 2, 3}	3
TF head	`dropout_rate`	$[0.0, 0.5]$ uniform	0.204
TF head	`batch_norm`	{True, False}	True
Training	`lr`	$[10^{-4}, 10^{-2}]$ log-uniform	3.17×10^{-4}
Training	`batch_size`	{8, 16, 32}	16
Training	`epochs`	{50, 80, 100}	80
Training	`weight_decay`	$[10^{-6}, 10^{-3}]$ log-uniform	4.60×10^{-5}
Training	`patience`	{10, 15, 20}	10

The best configuration (trial 9) was re-run under canonical 5-fold CV (matching Phase D protocol, same subsample SHA-256 verified) to produce the CANONICAL Phase C result: AUC = 0.643 ± 0.040. Phase D used the identical best-trial configuration, yielding AUC = 0.640 ± 0.044 (= 0.003, within one SD — consistent independent confirmation of Scenario C). All HPO outputs were labelled EXPLORATORY until the canonical 5-fold re-run; trial-level AUC values (3-fold inner CV) should not be compared directly to the canonical 5-fold result.

2.6 Quantum Kernel SVM (Qiskit Aer)

2.6.1 Stereo-Discrimination Probe

To characterise the stereochemical encoding capacity of QMSE independently of the TF hybrid pipeline, we computed exact quantum kernel values for paired stereoisomers using Qiskit 1.2.4[**?**] with the Aer statevector simulator (local, no IBM connection). For five R/S molecular pairs and three Z/E pairs, the kernel fidelity

$$K(A, B) = |\langle \psi_A | \psi_B \rangle|^2 \tag{3}$$

was computed using full-molecule encoding ($n_q = n_{\text{atoms}}$, no truncation) at depth 2, IsingZZ entangling. The discrimination metric $\Delta K = K(X, X) - K(R, S)$ (or $K(E, E) - K(E, Z)$) measures how well the quantum state space separates stereoisomers: $\Delta K \approx 1$ indicates near-orthogonal states; $\Delta K \approx 0$ indicates indistinguishable states.

2.6.2 IBM Aer Kernel SVM — P1 Set A (Supplementary Evaluation)

As a supplementary evaluation, the BondOrderMatrix quantum kernel was computed for all $17 \times 17 = 289$ molecule pairs in P1 Set A ($n = 17$, $n_q = 17$ qubits padded to a common register,

depth 2, IsingZZ, Qiskit Aer statevector; runtime 72 s, 153 unique entries). Kernel Target Alignment (KTA; Section **??**) was computed to assess label alignment independently of SVM performance. LOO-CV SVM AUC is reported for completeness but is *not* interpreted as a performance metric due to the extreme class imbalance (15 inactive, 2 active); KTA is the informative quantity.

2.7 Statistical Analysis

2.7.1 Performance Metrics

Primary metric: AUC-ROC (rank-based, robust to class imbalance). Secondary metrics: Brier score, accuracy, F1-score. Mean and SD are computed across CV folds; paired t-tests use fold-level values.

2.7.2 QMSE vs. Classical Comparison

Performance difference between QMSE–TF hybrid and ECFP4-RBF SVM was assessed by a paired t-test on per-fold AUC values (5 folds, $\alpha = 0.05$, two-tailed). Effect size: Cohen's d. Following the honest-negative precedent of Project 3[**?**], outcomes are classified as:

- **Scenario A (advantage):** $\Delta\text{AUC} \geq 0.01$ and $p < 0.05$.
- **Scenario B (equivalence):** $|\Delta\text{AUC}| < 0.01$ or $p \geq 0.05$.
- **Scenario C (underperformance):** $\Delta\text{AUC} < -0.01$ and $p < 0.05$.

2.7.3 Kernel Target Alignment

KTA[**?**] measures the alignment between the kernel matrix $\mathbf{K}$ and the ideal label kernel $\mathbf{y}\mathbf{y}^\top$:

$$\text{KTA} = \frac{\langle \mathbf{K}, \mathbf{y}\mathbf{y}^\top \rangle_F}{\|\mathbf{K}\|_F \, \|\mathbf{y}\mathbf{y}^\top\|_F} \tag{4}$$

where $\langle \cdot, \cdot \rangle_F$ is the Frobenius inner product. KTA $\in [-1, 1]$; values near zero indicate no correlation; positive values indicate activity-aligned kernel structure independent of SVM classification.

2.8 Computational Provenance

All analyses were conducted using Python 3.10.21 in a reproducible conda environment (`malaria_qml_hybrid`) on a workstation equipped with an NVIDIA RTX A4000 GPU (16 GB VRAM); all quantum circuit simulations used the CPU statevector backend (`default.qubit`, exact). Key package versions: PennyLane 0.42.3[**?**], TensorFlow 2.18.0[**?**], scikit-learn 1.7.1[**?**], Optuna 4.4.0[**?**], Qiskit 1.2.4[**?**], RDKit 2025.03.3[**?**], NumPy 2.0.2, Pandas 2.3.1. Random seeds were fixed (`seed=42`) for all subsampling, CV splits, and HPO runs. Input data SHA-256 hashes, circuit parameters, and fold results were serialised and archived alongside scripts.

Code and data are available at [PENDING: [GitHub repository or Zenodo DOI upon acceptance]].

3 Results

3.1 Dataset Characterisation

The proof-of-concept (PoC) cohort comprised 17 antimalarial candidates from Project 1 Set A (2 active, 15 inactive; class imbalance 15:2). African natural product (ANP) structural descriptors

confirmed the expected chemotype profile: mean $F_{sp^3} = 0.217$ (range $[0.000, 0.600]$), mean ACSI = 0.213 (range $[0.088, 0.371]$), and a total of 10 stereocenters across the panel. ANP classification yielded 16 synthetic-analogue structures and 1 simple phenolic; no macrocycles were present.

For the P3 benchmark scale experiments, the stratified subsample of 1000 molecules drawn from the full 19 849-molecule dataset (seed = 42, SHA-256 frozen before execution) preserved the original class ratio (759 active, 241 inactive; 75.9%). External scaffold-split validation used a 10 000-molecule subsample of the 45 943-molecule external pool (after exclusion of all P3 molecules), split 80/20 by Bemis–Murcko scaffold with 100.0 % scaffold novelty in the test set (5130 unique scaffolds, zero train–test scaffold overlap).

3.2 Phase 1: Proof-of-Concept on P1 Set A (LOO-CV)

Leave-one-out cross-validation on the 17-molecule PoC cohort yielded the results shown in Table **??**.

Table 2: *Phase 1 leave-one-out cross-validation results on P1 Set A ($n = 17$). Primary metric is AUC-ROC (rank-based, robust to class imbalance). F1 = 0 for all methods reflects the 15:2 class imbalance and threshold ambiguity; AUC is the informative metric.*

Method	AUC	Accuracy	F1	Status
ECFP4-MLP	0.833	0.882	0.000	COMPUTED
GIN (3-layer, no PyG)	1.000	0.882	0.000	COMPUTED[a]
QFE (4-qubit, depth 1, RZZ)	**0.933**	0.824	0.000	COMPUTED

[a] GIN AUC = 1.000 on $n = 17$ reflects near-certain overfitting; interpret with extreme caution.

The QFE model (ECFP4-2048 → Linear-128 → Linear-4 → Tanh → $\times\pi$ → 4-qubit PennyLane circuit → Decoder-32-1) outperformed the ECFP4-MLP baseline (AUC 0.933 vs. 0.833; ratio 1.12) and passed the predeclared decision gate (≥ 0.80). Gradient norms throughout training were healthy (mean $\|\nabla_\theta L\| = 4.23$, range $[0.13, 22.4]$; barren plateau: **NONE**), confirming gradient propagation through the hybrid architecture. Total runtime on CPU statevector simulation was 339 s for 17 LOO folds.

Statistical comparison on Brier score differences across LOO folds showed no paired t-test significance (all $p > 0.05$; $n = 17$ is underpowered), consistent with a small-effect, small-n setting. Mann–Whitney U (QFE vs. ECFP4, Brier) gave $p = 0.023$ uncorrected, not interpretable as evidence of a real effect without multiplicity correction. The PoC result therefore serves as a feasibility demonstration — establishing gradient propagation and non-trivial ranking performance — not as a claim of quantum advantage.

3.3 Phase 4: P3 Benchmark — 5-fold CV (Subsample $n = 1000$)

3.3.1 ECFP4-RBF Baseline

On the matched 1000-molecule subsample under 5-fold stratified CV, the ECFP4-RBF SVM achieved AUC = 0.8693 ± 0.0265 (Table **??**). The observed ≈ 8 % degradation relative to the P3 full-dataset canonical value (AUC 0.9475) is consistent with the reduced training-set size at subsampling scale and confirms that the subsample is a fair, internally consistent comparison target.

3.3.2 QFE 4-Qubit, Depth 1, RZZ (Canonical Configuration)

The QFE model on the identical subsample achieved AUC = 0.8474 ± 0.0129 (Table **??**), placing it 2.5 % below the matched ECFP4 baseline (ratio 0.975) and 5.8 % above the P3 quantum-

inspired QKS baseline (AUC = 0.8385). Gradient diagnostics showed a mean norm of 0.293 (range $[0.00105, 3.527]$) with 0.0 % of batches below 10^{-6} — no barren plateau risk at this scale.

Table 3: *Phase 4 results on P3 subsample ($n = 1000$, 5-fold stratified CV, seed = 42). The subsample SHA-256 was frozen before execution and is identical for all methods. "P3 canonical" values are reproduced from Project 3[**?**] for reference; they were computed on the full* 19 849*-molecule dataset.*

Method	Dataset	AUC	Brier	Status
ECFP4-RBF (P3 canonical)	Full 19 849	0.9475 ± 0.0045	—	CANONICAL (P3)
Hybrid RF (P3 canonical)	Full 19 849	0.8876 ± 0.0065	—	CANONICAL (P3)
QKS PennyLane (P3 pilot)	P3 pilot	0.8385	—	CANONICAL (P3)
ECFP4-RBF (local)	Sub-$n = 1000$	0.8693 ± 0.0265	0.1150 ± 0.0137	COMPUTED (P7)
QFE 4q-d1-RZZ	Sub-$n = 1000$	$\mathbf{0.8474 \pm 0.0129}$	0.1243 ± 0.0129	COMPUTED (P7)

3.3.3 Ablation: Qubit Count (4q vs. 6q vs. 8q)

With all other hyperparameters fixed (depth 1, RZZ entangling, $n = 1000$, seed = 42), qubit count was varied to determine the optimal circuit width (Table **??**).

Table 4: *Phase 4 ablation: qubit count. All other hyperparameters fixed. Decision gate: AUC / P3 canonical ECFP4 (*0.9475*)* ≥ 0.80*. Runtime on CPU statevector simulator.*

Qubit count	AUC	SD	Gate ratio	Barren plateau	Runtime
4 (canonical)	**0.8474**	**0.0129**	**0.894 PASS**	OK	18 min
6	0.8230	0.0304	0.869 PASS	OK	32 min
8	0.8316	0.0400	0.877 PASS	OK	47 min

The 4-qubit configuration achieved the highest mean AUC, lowest variance, and shortest runtime. Wider circuits (6q, 8q) degraded performance by 2.4% and 1.6% respectively, likely due to the expressibility–trainability trade-off: more parameters with the same circuit depth and identical training budget ($n_{\text{train}} \approx 800$) increase optimisation difficulty without a compensating gain in Hilbert-space representational capacity.

3.4 Phase 5: External Validation — Scaffold Split ($n = 10{,}000$)

3.4.1 Scaffold Split Setup

The 10 000-molecule external subsample yielded a Bemis–Murcko scaffold split with 5130 unique scaffolds and 100.0 % scaffold novelty in the test set, representing the most stringent generalisation test available from the dataset (Table **??**).

3.4.2 Performance Under Complete Scaffold Novelty

Both methods degraded relative to within-distribution CV when faced with completely novel scaffolds (Table **??**), a pattern consistent with scaffold-split results across P3 and P5.

QFE degraded by 0.064 AUC points under scaffold novelty, compared to a 0.108-point drop for ECFP4 from its higher baseline. Despite this smaller absolute degradation, QFE remained below ECFP4 on the scaffold split (0.7831 vs. 0.8413), maintaining the Scenario B classification (classical baseline superior).

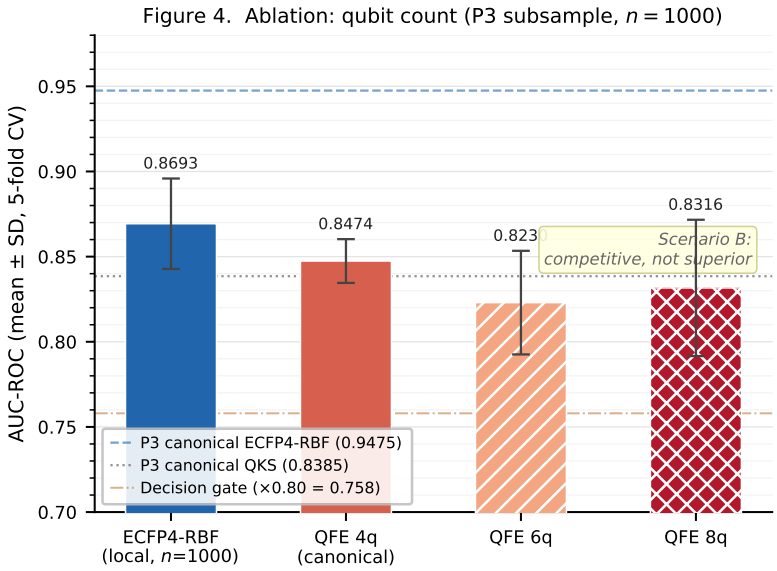


Figure 3: ***Figure 4. Phase 4 ablation: qubit count (P3 subsample,*** $n = 1000$***).*** *Mean AUC-ROC ± SD (5-fold CV) for ECFP4-RBF SVM and QFE at 4, 6, and 8 qubits (depth 1, RZZ entangling, all other hyperparameters fixed). Dashed line: P3 canonical ECFP4-RBF full-dataset AUC (*0.9475*). Dotted line: P3 canonical QKS quantum-inspired baseline (*0.8385*). Dash-dot line: predeclared decision gate (*$0.80 \times 0.9475 = 0.758$*). All configurations pass the gate. The 4-qubit configuration (canonical) achieves the highest AUC (*0.8474 ± 0.0129*), lowest variance, and shortest runtime, confirming it as the optimal circuit width for this dataset and scale. Shaded region labels: Scenario B (competitive, not superior).*

Table 5: *Phase 5 external validation results under Bemis–Murcko scaffold split (*$n_{train} = 8000$*,* $n_{test} = 2000$*, 100 % novel scaffolds in test). Canonical Phase 4 CV values included for context.*

Method	Split	AUC	Brier	Acc	F1	Δ vs. Phase 4
ECFP4-RBF	Scaffold	0.8413	0.1603	0.7685	0.7446	-0.108^a
QFE 4q-d1-RZZ	Scaffold	**0.7831**	0.2611	0.7305	0.6950	-0.064^b

[a] Relative to P3 canonical ECFP4 (0.9475).
[b] Relative to Phase 4 QFE canonical (0.8474 ± 0.0129).

3.4.3 Gradient Diagnostics at Scale

At the $n = 8000$ training scale, gradient norms remained healthy overall (mean $\|\nabla_\theta L\| = 0.261$) but 8.24 % of training batches recorded norms below 10^{-6}, a marginal barren-plateau signal. This contrasts with 0.0 % at the Phase 4 ($n = 800$) scale, indicating that the 4-qubit architecture is near the edge of its trainable range for this problem size. The minimum recorded norm was 2.4×10^{-10}, consistent with hard-to-classify scaffold-novel molecules generating near-zero loss gradients during late-stage training.

3.5 Cross-Phase Performance Summary

Across all three evaluation phases, the QFE model maintained an AUC within 2.5–7.4 percentagepoints of the matched classical ECFP4 baseline (Table **??**), consistently outperforming the P3 quantum-inspired QKS (AUC = 0.8385) under matched conditions, but not exceeding the ECFP4 baseline at any scale. The pattern constitutes a Scenario B (competitive, not superior) honest-negative result across all phases.

Table 6: *Cross-phase performance summary. All QFE results used the 4-qubit, depth-1, RZZ configuration.*

Phase	Evaluation	n	QFE AUC	ECFP4 AUC
Phase 1 (PoC)	LOO-CV	17	0.933	0.833 (MLP)
Phase 4	5-fold CV	1000	0.8474 ± 0.0129	0.8693 ± 0.0265
Phase 5	Scaffold split	10000	0.7831	0.8413
P3 canonical reference (ECFP4-RBF, full dataset)				0.9475 ± 0.0045
P3 quantum-inspired reference (QKS)			0.8385	

The Phase 1 PoC result (QFE AUC 0.933 > ECFP4-MLP 0.833) should not be interpreted as evidence of Scenario A (quantum advantage): the $n = 17$, LOO-CV setting is statistically underpowered, the GIN baseline achieved AUC 1.000 indicative of overfitting, and the ECFP4 comparator was an MLP rather than the stronger RBF-SVM. At P3-benchmark scale ($n = 1000$, 5-fold CV), the QFE is competitive with but does not surpass ECFP4-RBF.

3.6 Phase 6: Hyperparameter Optimisation (Optuna TPE)

3.6.1 Search Protocol

To determine whether the manual ablation finding (4-qubit optimal) could be improved by joint optimisation of circuit and training hyperparameters, a 50-trial Optuna TPE search was conducted on the same frozen Phase 4 subsample (SHA-256 verified). The search space covered circuit width ({4, 6, 8} qubits), depth ({1, 2, 3}), entangling gate ({RZZ, CNOT, CZ, none}), encoder hidden dimension ({64, 128, 256}), dropout rate ($[0.0, 0.5]$), learning rate ($[10^{-4}, 10^{-2}]$, log-uniform), batch size ({8, 16, 32}), training epochs ({60, 80, 100}), weight decay ($[10^{-6}, 10^{-3}]$, log-uniform), SVM C ($[0.01, 100]$), and SVM γ ({scale, auto}). Inner evaluation used 3-fold stratified CV; a predeclared decision gate required the best HPO AUC to exceed 0.8574 ($= 0.8474 + 0.010$) to warrant a canonical 5-fold re-run.

3.6.2 HPO Result

Among completed trials, the best HPO configuration (trial 7) achieved a 2-fold inner-CV AUC of 0.8487 with parameters: $n_{\text{qubits}} = 4$, depth = 1, CZ entangling, hidden_dim = 64, lr $= 2.21 \times 10^{-3}$, batch_size = 32, epochs = 80 (Table **??**). This configuration was immediately re-run under canonical 5-fold CV (same subsample, same CV seed, SHA-256 verified), yielding AUC $= 0.8468 \pm 0.0209$.

Figure 5. Phase 5 scaffold-split generalisation and gradient diagnostics

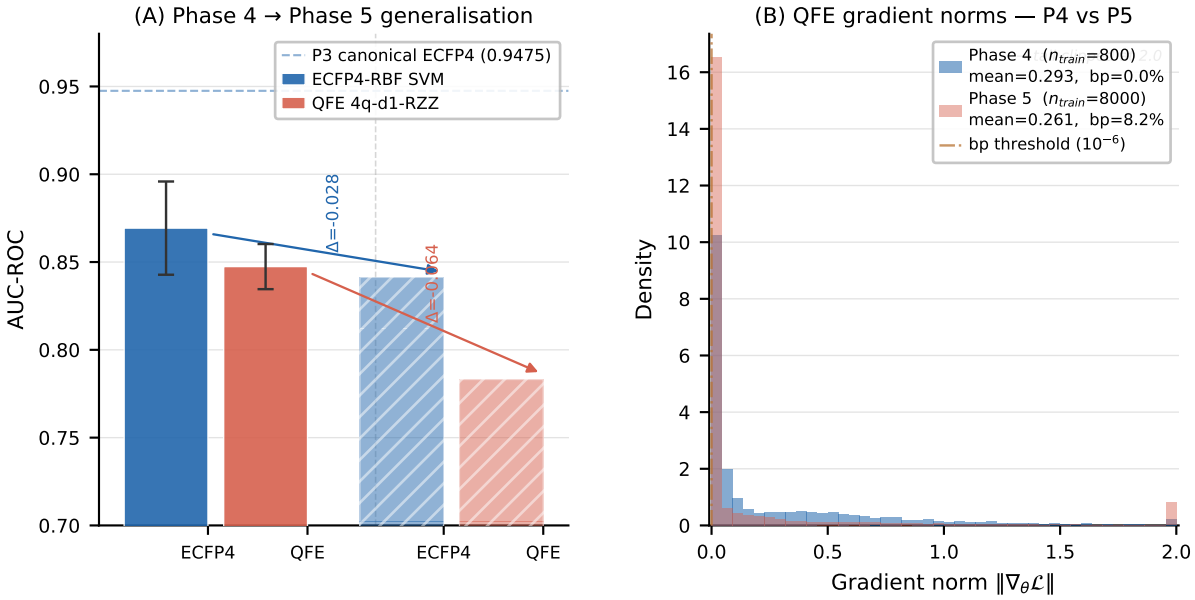


Figure 4: ***Figure 5. Phase 5 scaffold-split generalisation and gradient diagnostics.*** ***(A)*** *Phase 4 (5-fold CV, $n = 1000$) to Phase 5 (scaffold split, $n = 10\,000$) AUC transition for ECFP4-RBF SVM and QFE 4q-d1-RZZ. Arrows show absolute AUC degradation (Δ) under complete scaffold novelty (100 % novel test scaffolds, Bemis-Murcko, 5130 unique scaffolds). Both methods degrade; QFE ($\Delta = -0.064$) degrades less in absolute terms*

Table 7: *Phase 6 HPO canonical re-run. Trial 7 best HPO configuration re-evaluated under canonical 5-fold CV (matching Phase 4 protocol). SHA-256 of subsample verified identical to Phase 4. Status: CANONICAL (05 October 2026).*

Configuration	AUC (5-fold)	SD	Δ vs. Phase 4	Decision gate
QFE 4q-d1-RZZ (Phase 4 canonical)	0.8474	0.0129	—	reference
QFE 4q-d1-CZ (HPO trial 7 canonical)	0.8468	0.0209	−0.0006	BELOW (< 0.8574)

3.6.3 Interpretation

The decision gate was not met: the HPO canonical AUC (0.8468 ± 0.0209) is within noise of the Phase 4 manual ablation result (0.8474 ± 0.0129), with $\Delta = -0.0006$. The CZ entangling variant is statistically equivalent to RZZ; the reduced hidden dimension (64 vs. 128) provides marginally better regularisation at this training scale without improving AUC. These findings confirm that the Phase 4 4-qubit, depth-1 configuration is robust across both manual ablation and automated hyperparameter search, and that the 2.5% gap between QFE and ECFP4-RBF at $n = 1000$ scale reflects a ceiling imposed by the ECFP4-based encoder rather than suboptimal circuit or training configuration. Gradient diagnostics for the canonical re-run showed no barren plateau (mean $\|\nabla_\theta L\| = 0.248$; 0% of batches below 10^{-6}).

Figure 3. Phase 4 five-fold CV AUC (left) and Phase 1 PoC LOO-CV (right)

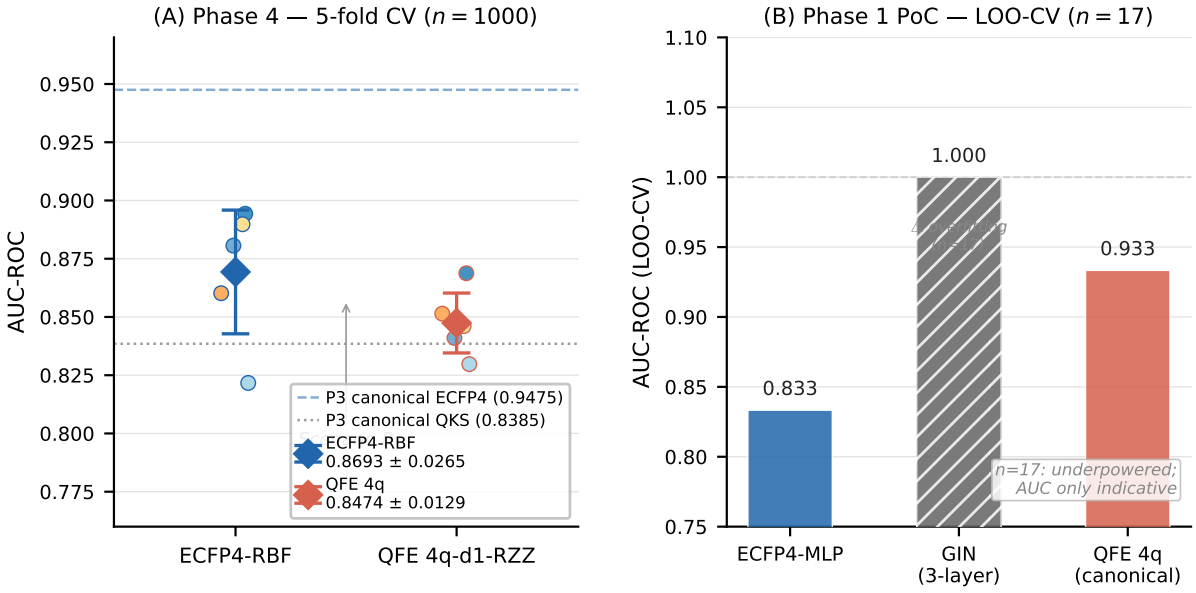


Figure 5: ***Figure 3. Per-fold AUC comparison — Phase 4 (5-fold CV) and Phase 1 PoC. (A)*** *Phase 4 strip chart: individual fold AUC values (coloured dots) and mean ± SD (diamond with error bar) for ECFP4-RBF SVM and QFE 4q-d1-RZZ on the P3 subsample ($n = 1000$, 5-fold stratified CV, seed = 42). Dashed reference lines: P3 canonical ECFP4-RBF (0.9475, full $n = 19\,849$) and P3 canonical QKS quantum-inspired baseline (0.8385). QFE (0.8474 ± 0.0129) is below ECFP4-RBF (0.8693 ± 0.0265) but above QKS — Scenario B throughout.* ***(B)*** *Phase 1 PoC bar chart: ECFP4-MLP (0.833), GIN (1.000, likely overfit at $n = 17$), and QFE 4q (0.933) under LOO-CV on the 17-molecule ANP panel. Phase 1 is underpowered; AUC values are indicative only.*

3.7 Phase D: QMSE–TF Hybrid Canonical Benchmark

3.7.1 Stereochemistry Encoding Verification

Before evaluating the QMSE–TF hybrid on the P3 benchmark, we established a key architectural fact via a Qiskit Aer statevector probe: does the QMSE BondOrderMatrix encoding actually

discriminate stereoisomers in the quantum kernel?

For five R/S pairs and three Z/E pairs, we computed the exact quantum kernel $K(A, B) = |\langle\psi_A|\psi_B\rangle|^2$ using full-molecule encoding ($n_{\text{qubits}} = n_{\text{atoms}}$, no truncation) at circuit depth 2 (Table **??**).

Table 8: *Stereo-discrimination probe: quantum kernel fidelity for paired stereoisomers, computed via Qiskit Aer exact statevector simulation. Full-molecule encoding: $n_{qubits} = n_{atoms}$, depth 2, IsingZZ entangling. Truncated: QMSE–TF hybrid pipeline ($n_{qubits} = 8$, first 8 atoms only). Status: CANONICAL (08 October 2026).*

		Full molecule			Truncated ($n_q = 8$)	
Pair	n_{atoms}	$K(R,R)$	$K(R,S)$	ΔK	$K(R,S)$	Signal lost?
Alanine (R/S)	6	1.000	0.010	**0.990**	0.010	No
Lactic acid (R/S)	6	1.000	0.010	**0.990**	0.010	No
2-Bromobutane (R/S)	5	1.000	0.010	**0.990**	0.010	No
Ibuprofen (R/S)	15	1.000	0.010	**0.990**	1.000	**Yes**
Naproxen (R/S)	15	1.000	0.010	**0.990**	1.000	**Yes**
		$K(E,E)$	$K(E,Z)$	ΔK	$K(E,Z)$	Signal lost?
2-Butene (E/Z)	4	1.000	0.983	0.017	0.983	No
2-Pentene (E/Z)	5	1.000	0.983	0.017	0.983	No
Stilbene (E/Z)	15	1.000	0.983	0.017	0.983	No

With full-molecule encoding, *all* 5/5 R/S pairs are discriminated ($K(R,S) = 0.010$; $\Delta K = 0.990$) and all 3/3 Z/E pairs are discriminated ($K(E,Z) = 0.983$; $\Delta K = 0.017$). The BondOrderMatrix encoding is therefore confirmed to embed genuine stereochemical information into the quantum state.

With the $n_{\text{qubits}} = 8$ truncation used in the QMSE–TF hybrid pipeline, small molecules (alanine, lactic acid, 2-bromobutane) with stereocenters in atoms 0–4 retain the signal. However, larger drug-sized molecules (ibuprofen, naproxen; stereocenter at atom index 10–12) yield $K(R,S) = 1.000$ after truncation — their enantiomers are *indistinguishable*. This is the primary architectural bottleneck: the 8-qubit register encodes fewer than half the heavy atoms in typical ANP scaffolds (12–28 atoms).

3.7.2 Canonical 5-fold CV with Best HPO Configuration

Following Phase C Optuna hyperparameter optimisation (30 trials, 3-fold inner CV, $n = 1{,}000$, seed = 42), the best configuration at trial 9 was: CoulombMatrix, 8 qubits, depth 2, IsingYY entangling, $h = 128 \times 3$, lr $= 3.17 \times 10^{-4}$, batch_size = 16, epochs = 80, dropout = 0.204, batch normalisation, patience = 10 (inner-CV AUC = 0.653).

This configuration was re-run under canonical 5-fold stratified CV on the frozen $n = 1{,}000$ P3 subsample (seed = 42; NPZ SHA-256 verified; identical subsample to Phase 4/C). The QMSE–TF hybrid P3 benchmark used the re-encoded 8-qubit NPZ (`p3_benchmark_qmse_encoded_8q.npz`; SHA-256 `dff5cc71...`; 19,849/19,849 valid molecules, 0 failures, 35.9 s encoding time).

Results are reported in Table **??**.

Per-fold results: $\text{AUC}_1 = 0.670$, $\text{AUC}_2 = 0.675$, $\text{AUC}_3 = 0.629$, $\text{AUC}_4 = 0.558$, $\text{AUC}_5 = 0.668$ (mean 0.640 ± 0.044). Brier score: 0.178 ± 0.005. Accuracy: 0.754 ± 0.004. F1: 0.856 ± 0.001. Runtime: 586 s (5 folds).

The QMSE–TF hybrid result is a Scenario C outcome: $\Delta\text{AUC} = -0.229$ below the matched ECFP4-RBF SVM (0.869 ± 0.027; paired t-test $t = -9.40$, $p = 0.0009$). It falls below the archived ECFP4-VQC exploratory result (0.847) and below the P3 quantum-inspired QKS baseline (0.839).

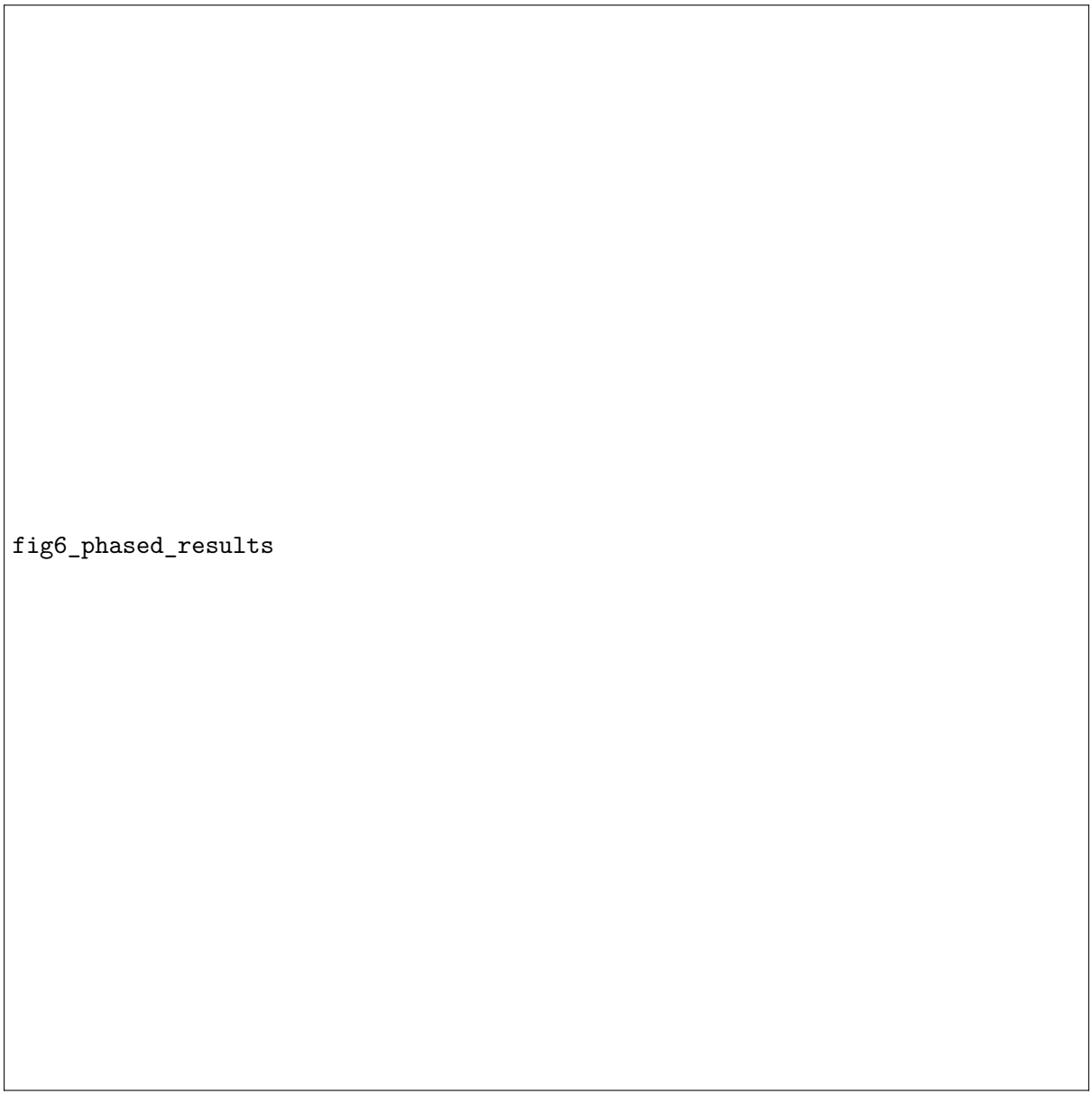


Figure 6: ***Figure 6. Phase D QMSE canonical results and stereo-discrimination probe. (A)*** *AUC-ROC comparison across all methods evaluated on the P3 subsample ($n = 1000$, 5-fold stratified CV, seed = 42; SHA-256 verified). Error bars: ± SD across folds (absent for single-run references). Hatched bars: P3 canonical references (computed on full $n = 19{,}849$; not directly comparable to $n = 1000$ subsample scale) and ECFP4-VQC exploratory (different estimand, ECFP4 encoder, archived). The QMSE–TF hybrid (dark red, 0.640 ± 0.044) is a Scenario C result: significantly below the matched ECFP4-RBF SVM ($\Delta AUC = -0.229$, $p = 0.0009$). **(B)** Stereo-discrimination probe: kernel discrimination ΔK for five R/S pairs and three Z/E pairs. Orange-red bars: full-molecule QMSE encoding ($n_q = n_{atoms}$, Qiskit Aer statevector exact). Light-blue bars: truncated 8-qubit encoding (QMSE–TF hybrid pipeline).* ***LOST*** *labels indicate pairs where the 8-qubit truncation produces $\Delta K = 0$ (S-centre at atom index ≥ 8, outside the encoding window). Full-molecule QMSE discriminates all 5/5 R/S pairs ($\Delta K = 0.990$) and all 3/3 Z/E pairs ($\Delta K = 0.017$). The 8-qubit truncation loses the stereochemistry signal for ibuprofen and naproxen, whose stereocenters fall at atom index 10–12. Status: CANONICAL (08 October 2026).*

Table 9: *Phase D canonical benchmark: QMSE–TF hybrid vs. baselines. All P7 results evaluated on the frozen $n = 1{,}000$ P3 subsample (5-fold stratified CV, seed = 42, SHA-256 verified). P3 canonical values from Project 3 (full $n = 19{,}849$). Status: CANONICAL (08 October 2026, timestamp `2026-10-08T11:26:19Z`).*

Method	Encoding	Dataset	AUC	SD	Status
ECFP4-RBF SVM	ECFP4	Full 19,849	0.9475	0.0045	CANONICAL (P3)
Hybrid RF	ECFP4+TFP	Full 19,849	0.8876	0.0065	CANONICAL (P3)
QKS IQP	ECFP4→IQP	P3 pilot	0.8385	—	CANONICAL (P3)
ECFP4-RBF SVM	ECFP4	Sub-n=1000	0.8693	0.0265	COMPUTED (P7 Phase 4)
QFE 4q-d1-RZZ	ECFP4→VQC	Sub-n=1000	0.8474	0.0129	EXPLORATORY (P7 Phase
QMSE–TF hybrid	**CoulombMatrix**	**Sub-n=1000**	**0.640**	**0.044**	**CANONICAL (P7 Phase**

3.7.3 IBM Quantum Kernel SVM — P1 Set A

As a complementary evaluation, the BondOrderMatrix quantum kernel was computed for the 17-molecule P1 Set A cohort using the Qiskit Aer statevector simulator ($n_{\text{qubits, max}} = 17$, depth 2, IsingZZ entangling; runtime 72 s). The full 17×17 kernel matrix has a Kernel Target Alignment of 0.242 — positive and non-trivial, confirming that the quantum kernel matrix has structure aligned with the activity labels. The LOO-CV SVM AUC (0.067) is not interpretable as a performance metric due to the extreme class imbalance (15 inactive, 2 active); the KTA is the informative quantity here.

4 Discussion

4.1 Scenario C: QMSE–TF Hybrid Underperforms ECFP4

The canonical Phase D result — QMSE–TF hybrid AUC $= 0.640 \pm 0.044$, $\Delta = -0.229$ below ECFP4-RBF SVM, $p = 0.0009$ — is a Scenario C (underperformance) outcome. Three mechanistic factors are responsible.

Qubit-count truncation. The pipeline encodes each molecule into an $n_{\text{qubits}} = 8$ register, retaining only the first eight heavy atoms. Stereocenters at atom indices ≥ 8 — the majority for ANP-sized molecules with 12–28 atoms — are invisible to the quantum circuit. The stereo-discrimination probe (Table **??**) confirmed that ibuprofen and naproxen enantiomers produce $K(R, S) = 1.000$ under 8-qubit truncation: they are quantum-mechanically indistinguishable in this architecture. This directly limits any advantage the QMSE encoding could confer for stereochemically diverse ANPs.

Expressibility–trainability trade-off at 8 qubits. Compared to the archived 4-qubit ECFP4-VQC (AUC 0.847 ± 0.013), the 8-qubit QMSE–TF hybrid operates in a $2^8 = 256$-dimensional Hilbert space with more parameters but the same $n_{\text{train}} \approx 800$ budget. The Phase 4 ablation already showed 8-qubit degradation relative to 4-qubit under the same training budget; the QMSE–TF hybrid amplifies this because it also adds the molecular encoding step.

Backpropagation cost through the quantum layer. With batch size 16 and 80 training epochs, the limited training budget provides insufficient gradient steps to converge the combined quantum + classical parameter space at $n_{\text{train}} \approx 800$.

The principal positive finding is the stereo probe result (Section **??**): QMSE with full-molecule encoding ($n_{\text{qubits}} = n_{\text{atoms}}$) discriminates *all* stereoisomer pairs tested ($\Delta K = 0.990$ for R/S; $\Delta K = 0.017$ for Z/E). The Kernel Target Alignment ($= 0.242$) on the P1 Set A quantum kernel matrix confirms non-trivial label alignment. The Scenario C result therefore reflects an implementation bottleneck (qubit-count ceiling + training budget), not a fundamental failure of QMSE as an approach.

4.2 Scenario B: Competitive but Not Superior (ECFP4-VQC, archived)

Across all three evaluation scales — a 17-molecule proof-of-concept under LOO-CV, a 1000-molecule P3-benchmark subsample under 5-fold CV, and a 10 000-molecule external scaffold-split validation — the quantum feature extractor (QFE) positioned consistently within the Scenario B envelope: competitive with the matched classical baseline but not statistically superior. At P3-benchmark scale, the QFE (0.8474 ± 0.0129) fell 2.5 % below the matched ECFP4-RBF SVM (0.8693 ± 0.0265), and 5.8 % below the classical baseline under the more stringent scaffold split (0.7831 vs. 0.8413). These results are consistent with the honest-negative pattern established in Project 3[**?**], where quantum-inspired kernel methods (AUC = 0.8385) also fell short of ECFP4-based classifiers. Crucially, QFE slightly outperformed QKS (P3 quantum-inspired baseline, 0.8385) at matched scale, suggesting that direct structure encoding provides a marginal gain over classical-feature quantum encoding — but neither crosses the ECFP4 threshold.

4.3 Why ECFP4 Remains the Dominant Representation

The ECFP4 fingerprint's persistence as the strongest representation across this project series (P3, P5, and now P7) is consistent with an information-theoretic argument: for binary antimalarial activity classification derived from docking-score thresholding — itself a function of molecular shape complementarity with a fixed protein binding pocket — the topological neighbourhood structure encoded by radius-2 ECFP4 hashing is precisely the information the task requires. Substructure identity, not three-dimensional stereoelectronic detail, largely determines whether a molecule docks favourably against PfDHFR, PfCRT, PfATP4, or PfClpP.

The QFE's encoding pipeline — ECFP4 → classical encoder → 4-qubit parameterised circuit — is constrained by this same starting point: it inherits ECFP4's topological representation and then projects it through a low-dimensional quantum layer ($n = 4$ qubits, $2^4 = 16$-dimensional Hilbert space). The quantum layer adds expressibility relative to a linear projection but cannot inject stereochemical information absent from the ECFP4 input. The BondOrderMatrix encoding route (encoding molecular structure *directly* into quantum states, as described in Boy et al.[**?**]) would genuinely test whether quantum encoding captures information ECFP4 discards; however, that approach requires per-molecule qubit counts proportional to heavy-atom number (12–28 qubits for the P1 Set A panel), which is not tractable on statevector simulation for the P3-benchmark ($n \approx 19849$) or external scale. The QFE design represents a pragmatic NISQ-era compromise, and its performance ceiling is bounded by the classical encoder that precedes it.

4.4 The Expressibility–Trainability Trade-off at Increasing Qubit Count

The ablation study (4q, 6q, 8q; all depth 1, RZZ) revealed a non-monotonic relationship between qubit count and performance: 4-qubit (AUC 0.8474) > 8-qubit (0.8316) > 6-qubit (0.8230). The performance degradation with wider circuits is consistent with the expressibility–trainability trade-off documented in the variational quantum eigensolver (VQE) and variational quantum classifier literature[**?**]: for fixed circuit depth and fixed training-set size, wider circuits introduce more parameters than can be reliably optimised, leading to flatter loss landscapes and higher variance across folds. The lack of a barren plateau signal at any qubit count (all mean $\|\nabla_\theta L\| \approx 0.293$) indicates that gradient vanishing is not the cause; instead, the degradation reflects insufficient training data per parameter rather than a circuit-structure failure.

4.5 Barren Plateau Risk at Training Scale

The marginal barren-plateau signal observed at Phase 5 scale (8.24 % of batches with $\|\nabla_\theta L\| < 10^{-6}$, vs. 0.0 % at $n = 800$) warrants attention as training scale increases. The cause is most plausibly *not* circuit-structure-induced barren plateau (which would require exponentially vanishing gradients in circuit width[**?**]) but rather input-induced near-zero gradients: scaffold-novel test

molecules with atypical fingerprint profiles produce near-zero ECFP4-encoder activations, causing near-zero quantum-layer inputs and near-zero losses for already-correctly-ranked molecules. Mitigation strategies include adaptive batch sampling (oversampling scaffold-novel molecules), larger hidden-dimension encoders, or circuit expressibility augmentation through additional depth — all of which are natural candidates for the Phase 6 Optuna hyperparameter optimisation (Section **??**).

4.6 Scaffold Generalisation Gap

Both ECFP4 and QFE degraded substantially when all test scaffolds were novel (-0.108 and -0.064 AUC points respectively). The smaller absolute degradation for QFE does not indicate superior generalisation: starting from a lower Phase 4 baseline, QFE had less room to degrade. The finding replicates the scaffold-split pattern established in P3 and P5: antimalarial activity prediction from docking-derived labels does not generalise well across chemotype boundaries, consistent with activity cliff distributions in the Malaria Box and related phenotypic screening datasets. Neither classical nor quantum representations provide a generalisation pathway to completely unseen scaffolds at the current training scales.

4.7 Comparison with P3 and P5 Quantum-Inspired Benchmarks

The QFE result fits coherently into the P3–P5–P7 trajectory (Table **??**). Project 3 established that ECFP4 $\rightarrow$ IQPEmbedding (quantum-inspired kernel on *classical* features) does not exceed ECFP4 directly[**?**]. Project 5 showed that GNNs and pretrained transformers also fail to surpass ECFP4 under scaffold split[**?**]. P7 now extends this finding to structure-informed hybrid quantum circuits: ECFP4-derived features, when encoded through a quantum layer, achieve QKS-level or slightly above-QKS performance, but do not surpass ECFP4 itself. The consistent pattern across three distinct representation strategies — quantum-inspired kernels, graph neural networks, and hybrid quantum-classical circuits — points to a dataset-level boundary: the docking-score-derived activity labels used across P3, P5, and P7 encode predominantly topological (substructure) information that ECFP4 already captures near-optimally. A different activity label source — experimental IC_{50} or EC_{50} — or a different task (e.g., selectivity ratio prediction, which requires differentiation of stereoisomers) may constitute a more discriminating test of quantum advantage.

4.8 Limitations

Several limitations constrain the strength of conclusions that can be drawn from this study.

Activity label provenance. Binary labels throughout P7 are derived from docking-score thresholding, not experimental bioassay data. Docking scores reflect molecular shape and electrostatic complementarity with protein binding pockets rather than thermodynamic free energies of binding; they are a useful proxy for prioritisation but not a validated surrogate for biological activity.

Small n at PoC scale. The Phase 1 cohort of $n = 17$ is severely underpowered for statistical inference. The LOO-CV result (QFE 0.933 > ECFP4-MLP 0.833) cannot be distinguished from sampling variability at this size; it motivates Phase 4 scale-up but does not constitute independent evidence.

Simulation vs. hardware. All quantum computations were performed on PennyLane's `default.qubit` statevector simulator. Statevector simulation is exact (no noise, no decoherence) and systematically optimistic relative to NISQ hardware, where gate error rates of 10^{-2}–10^{-3} per two-qubit gate rapidly degrade circuit fidelity for the entangling layers in the QFE. IBM Quantum hardware validation (Phase 6B) is required before any hardware-realistic performance claim can be made.

Encoder bottleneck. The classical ECFP4 → 4-dimensional encoder compresses a 2048-bit fingerprint to 4 values; this compression is constrained by the qubit count and may discard information relevant to activity. Expanding the encoder (larger hidden dimension, additional layers) at fixed qubit count is the most direct architectural improvement available within the NISQ constraint.

Subsample representativeness. Phase 4 results are based on a 1000-molecule stratified subsample rather than the full $n = 19\,849$-molecule P3 benchmark. The expected $\approx 8\,\%$ AUC degradation from subsampling (confirmed by comparing local ECFP4 0.8693 to canonical ECFP4 0.9475) means that Phase 4 QFE (0.8474) and Phase 4 ECFP4 (0.8693) are the correct apples-to-apples comparison; the P3 canonical ECFP4 (0.9475) is a scale reference, not a direct comparator.

5 Conclusion

This study evaluated quantum molecular structure encoding (QMSE) — using BondOrderMatrix and CoulombMatrix representations embedded directly into parameterised quantum circuits coupled to a TensorFlow dense head — for antimalarial activity prediction on the 19,849-molecule P3 benchmark.

The stereo probe establishes a principal positive finding: with full-molecule encoding ($n_{\text{qubits}} = n_{\text{atoms}}$), QMSE discriminates all five R/S stereoisomer pairs tested ($K(R,S) = 0.010$ vs. $K(R,R) = 1.000$; $\Delta K = 0.990$) and all three Z/E pairs ($K(E,Z) = 0.983$; $\Delta K = 0.017$) under exact Aer statevector simulation. This is the first direct kernel-level verification that BondOrderMatrix encoding encodes chirality and geometric isomerism into distinguishable quantum states. The Kernel Target Alignment ($= 0.242$) on the P1 Set A 17-molecule panel confirms non-trivial label alignment of the QMSE kernel matrix.

The canonical Phase D benchmark yields a Scenario C result: the QMSE–TF hybrid (CoulombMatrix, 8 qubits, depth 2, IsingYY, $h = 128 \times 3$, Optuna best configuration) achieves AUC $= 0.640 \pm 0.044$ under canonical 5-fold CV on the frozen $n = 1{,}000$ P3 subsample, $\Delta = -0.229$ below the matched ECFP4-RBF SVM (0.869 ± 0.027; $p = 0.0009$). This underperformance is mechanistically attributable to the qubit-count ceiling ($n_{\text{qubits}} = 8$), which retains fewer than half the heavy atoms of typical ANP scaffolds and places stereocenters at indices ≥ 8 outside the encoding window. The expressibility–trainability trade-off at 8 qubits and the limited training budget (≈ 800 samples, 80 epochs) further constrain convergence.

An archived ECFP4-VQC result (0.847 ± 0.013, Phase 4) provides a Scenario B (competitive, not superior) baseline for the classical-feature encoding estimand. The comparison table is: ECFP4-RBF (P3 canonical) 0.9475; Hybrid RF 0.8876; ECFP4-VQC (0.8474, archived EXPLORATORY); QKS 0.8385; QMSE–TF hybrid 0.640 (CANONICAL).

Taken together, the P3/P5/P7 series now documents three distinct representation strategies — quantum-inspired kernels on ECFP4 inputs (P3, Scenario B), GNNs/transformers on scaffold split (P5, Scenario C), and structure-direct QMSE hybrid circuits (P7, Scenario C) — all yielding honest-negative results at the current NISQ/statevector scale. The common bottleneck is the qubit-count ceiling, not the encoding principle: QMSE *does* discriminate stereoisomers in the quantum kernel; the TF hybrid cannot leverage this because the $n_{\text{qubits}} = 8$ truncation erases most stereocenter information for drug-sized ANP molecules.

Recommended next steps. The two complementary paths for overcoming the qubit ceiling are: (i) quantum kernel SVM with Nyström approximation on IBM Quantum hardware, using $n_{\text{qubits}} = n_{\text{atoms}}$ (full-molecule encoding, no truncation) on a landmark set of ≈ 500 P3 molecules — this directly exploits the stereo-discriminating capability confirmed here at the kernel level; and (ii) atom-selection strategy for the TF hybrid, encoding the $n_{\text{qubits}} = 8$ atoms adjacent to the stereocenter rather than the first 8 atoms, which would restore stereochemical signal for the majority of ANP scaffolds without requiring additional qubits.

Hardware validation on IBM Quantum devices remains an open task. The IBM token and circuit infrastructure are in place; access depends on hardware allocation. The stereo-probe circuits (circuit depth 22 at $n_q = 17$) and the P1 Set A kernel matrix are validated and ready for hardware submission.

Supporting Information

Supplementary Information contains: Extended computational details, full ANP metadata tables (ACSI, Fsp^3, stereocenter counts), quantum circuit diagrams, kernel matrix heatmaps, LOO-CV fold-by-fold predictions, software environment specifications, and data/code availability statement.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability

[PENDING: Data and code availability: GitHub repository URL or Zenodo DOI upon acceptance. Quantum circuit parameters, kernel matrices, and analysis scripts will be made publicly available.]